

Figure 1: SQL vs NoSQL alternatives

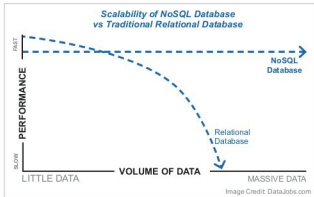


Figure 2: SQL vs NoSQL performance at scale

vehicle. What he means to say is that very few organisations (that handle large volumes of data like Google and Facebook) will actually be bound by the performance requirements that dictate a switch from SQL to NoSQL.

NoSQL technologies

Apache HBase

Apache HBase is an open source column-oriented NoSQL database that runs on top of HDFS, and is often used in cases where data access is required to be in real-time. Known as the 'Hadoop Database', it addresses problems pertaining to the manipulation of unstructured data, and offers significant functionality in areas such as scalability, failover support and sharding. The primary precondition of huge and sparse data sets (and we're talking of millions of rows) is what usually introduces HBase into the design of a system.

HBase has a number of features with a focus on write-optimisation, although scans are also pretty fast on the system. However, it is not meant to be a replacement for the traditional relational database management system (RDBMS). One of the factors to be noted while deploying an HBase cluster is that there must be monitoring of nodes, since failure is cascading and can potentially bring down the system.

Recent updates have introduced Yam integration within HBase, along with improved availability and support for

data types. Originally designed based on Google's BigTable to work on top of the Hadoop ecosystem, today HBase is used in production for real-time analytics at Facebook, for storage of graph data at Pinterest, and for personalisation of the content feed for users on Flipboard.

Apache Cassandra

In the conventional RDBMSs, replication and scaling were left to the users, which they discovered much to their dismay when large scale use cases came to the fore. Apache Cassandra presents a refreshing break from the configuration woes, going so far as to present its own query language, CQL, designed to be 'exactly like SQL, except when it's not'. Cassandra offers prebuilt support for multiple data centres, and all that the user is required to do is provide information about the other systems running it. For instance, adding a Cassandra node to a cluster can be as simple as booting a new machine, installing the software and telling it where the other nodes are.

Cassandra outdoes its competition in terms of ease of deployment, with the sole overhead being that of understanding the data model and how it will integrate with a given application. It does have the drawback of losing some of the performance on the introduction of a secondary indexing database. Offering lightning-fast write speeds and a predictable query performance makes Cassandra a great competitor in this segment.

MongoDB

The creators of MongoDB struck a compromise between its support for diverse data sets, its capacity to scale horizontally, and a functionality that is similar to that of a relational database. It is a great product to use out-of-the-box for a wide variety of applications in multiple scenarios. MongoDB is often one of the first preferences for developers due to how easy it is to understand and its gradual learning curve.

It does have its own share of issues, however, which adversely impact its suitability for reporting-style tasks in some cases, but it remains attractive for OLTP workloads. Secondary indexing, though, finds greater support in MongoDB than Cassandra, allowing for nested, complex queries to be executed. It does have a latency for failovers that is higher than Cassandra, but its ease of use outweighs this downside and makes MongoDB an extremely popular and widely employed NoSQL database. 

By: Swapneel Mehta

The author has worked with Microsoft Research, CERN and startups in the AI and cyber security domains. An open source enthusiast, he enjoys spending his time organising software development workshops for school and college students. You can connect with him at <https://www.linkedin.com/in/swapneelm> and find out more at <https://github.com/SwapneelM>.